

No. of Functional Features Included in the Solution

Date	02 October 2026
Team ID	6ab7f80ad083b0e2ee72b17e
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Question Answering	Answers a learner's question in simple language, explains key terms and says when something is uncertain (/qa).	qna.py	Done	High
2	Concept Explanation	Explains a topic at beginner, intermediate or advanced level using a short definition, an example and a recap (/explain).	explanation_module.py	Done	High
3	Interactive Quiz	Generates a 3-question multiple-choice quiz from a topic or passage, with instant correct / incorrect feedback (/quiz).	quiz_module.py	Done	High
4	Notes Summarization	Summarizes study notes or a passage into key points for quick revision (/summarize).	summary_module.py	Done	Medium
5	Learning Path	Creates an ordered learning path with prerequisites, time estimates and practice ideas (/learn/recommendations).	learning_path.py	Done	Medium
6	Web Interface	Single page with task and level selection, text input, formatted results, quiz feedback and a copy-result button.	templates/index.html, static/style.css	Done	High
7	AI Model Integration	Shared Gemini client used by all AI features; optional local LaMini-Flan-T5-783M for explanations with automatic Gemini fallback.	ai_client.py, explanation_module.py	Done	High
8	Validation, Error Handling and Health Check	Input length and empty-text validation (HTTP 422), clear error messages in the UI, and a /health endpoint.	main.py, ai_client.py	Done	Medium
9	Learning-Gap Detection	Diagnostic readiness check to show what the student understands and which topics need attention.	Not yet built	Pending	Planned
10	Study Planner	Study plan built from subjects, deadlines and available time.	Not yet built	Pending	Planned

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11	Progress Tracking	Goals, progress tracking and a dashboard that keeps context across sessions (needs storage).	Not yet built	Pending	Planned

Feature Summary:

Metric	Count / Value
Total Features Planned	11
Total Features Implemented	8
Core / Must-Have Features	7 (question answering, explanation, quiz, summarization, learning path, web interface, AI integration)
Additional / Nice-to-Have Features	4 (validation and health check, learning-gap detection, study planner, progress tracking)
Features Tested & Verified	8 (quiz JSON clean-up by unit tests; all endpoints checked manually through the browser and /docs)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Task and level selector, text input, interactive quiz feedback, copy-result button
2	Backend / Logic	6	Question answering, explanation, quiz generation, summarization, learning path, input validation
3	Database / Storage	0	No database in the prototype; needed later for progress tracking
4	API / Integration	1	Gemini API client, optional local LaMini-Flan-T5 model, REST endpoints (/qa, /explain, /quiz, /summarize, /learn/recommendations, /health)
5	Security / Authentication	0	API key kept in .env and input validation only; user authentication and rate limiting are pending